

MSE-Optimal Difference-in-Differences Estimator

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Overview (Flash Talk)

- Conventional difference-in-differences (DiD) models often face concerns about:
 - **Accuracy**
 - Estimates may be imprecise, especially in small samples.
 - **Validity**
 - Pre-trend tests may have low power to detect violations of parallel trends (Roth, 2022).
- Consequently, using **more pre-treatment periods** can **reduce variance** by using more pre-treatment data, but it may also **increase bias** if the parallel trends assumption is violated over a longer period.
- This creates a **bias–variance tradeoff**.
- I propose an **MSE-based framework** for choosing the **optimal length of pre-treatment periods** and evaluate its applicability using simulations and an empirical application.

Model

- Let $\ell \in \mathcal{L} = \{0, 1, \dots, L\}$ denote the length of pre-treatment periods. For each $\ell \in \mathcal{L}$, let $\hat{\beta}_\ell$ denote the corresponding DiD estimator.
- The goal is to choose ℓ to **minimize the mean squared error (MSE)** of the target DiD estimator:

$$\begin{aligned}\ell^* &= \arg \min_{\ell} \text{MSE}(\hat{\beta}_\ell) \\ &= \arg \min_{\ell} \left(\text{Bias}(\hat{\beta}_\ell)^2 + \text{Var}(\hat{\beta}_\ell) \right)\end{aligned}$$

- The variance is estimated by the squared standard error, while the bias is approximated by the deviation from the benchmark estimate with $\ell = 0$.

Model

- Specifically, in the **dynamic treatment effect** setting, I consider a modified event study specification that uses **all pre-treatment periods as reference periods** to improve precision.

$$Y_{it} = \mu_i + \theta_t + \sum_{\ell \geq 0} \beta_{\ell} \mathbf{1}\{t - t_i^* = \ell\} + \varepsilon_{it} \quad (1)$$

- Under this specification, the estimator of β_{ℓ} is

$$\begin{aligned} \hat{\beta}_{\ell}^m = & \frac{1}{\sum_{i=1}^N \mathbf{1}\{D_i = 1\}} \sum_{D_i=1} (Y_{i,t_i^*+\ell} - Y_{i,(t \leq t_i^*)}) \\ & - \frac{1}{\sum_{i=1}^N \mathbf{1}\{D_i = 0\}} \sum_{D_i=0} (Y_{i,t_i^*+\ell} - Y_{i,(t \leq t_i^*)}) \end{aligned}$$

Simulation

A simulation evaluates the performance of the MSE-optimal method in a **dynamic treatment effect** setting.

Table : Simulation results when the target parameter is β_5 ($n = 50$)

Length of Pre-trends	MSE	Bias Squared	Variance	Coefficient	SE
0	1017.981	0.000	1017.981	47.099	31.906
1	676.540	51.938	624.602	54.306	24.992
2	645.241	89.283	555.958	56.548	23.579
3	508.401	18.735	489.665	51.428	22.128
4	525.972	7.822	518.150	49.896	22.763
5	504.936	25.067	479.870	52.106	21.906
6	499.688	14.774	484.914	50.943	22.021
7	526.362	60.236	466.126	54.860	21.590
8	600.299	133.914	466.385	58.671	21.596
9	727.893	269.806	458.088	63.525	21.403

Simulation

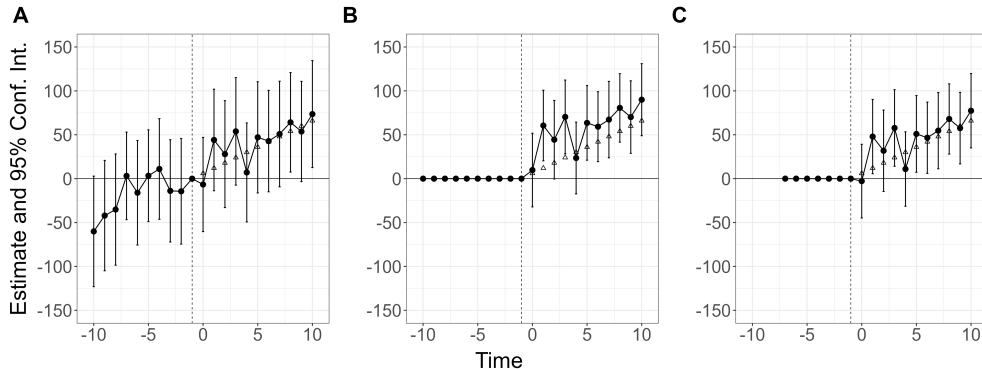


Figure : Event study plots across specifications

- (A) Conventional event study specification
- (B) Modified event study specification (1)
- (C) **MSE-optimal event study specification**

Empirical Illustration

I apply the MSE-optimal DiD framework to Fitzpatrick and Lovenheim (2014), who study the effect of Illinois's 1993 Early Retirement Incentives (ERI) program on student achievement.

Table : Estimates by length of pre-treatment periods

Length of Pre-trends	MSE	Bias Squared	Variance	Coefficient	SE
0	0.0000250	0.0000000	0.0000250	0.00982	0.00500
1	0.0000266	0.0000004	0.0000262	0.00917	0.00512
2	0.0000247	0.0000004	0.0000243	0.01042	0.00493
3	0.0000245	0.0000011	0.0000234	0.01086	0.00484

Empirical Illustration

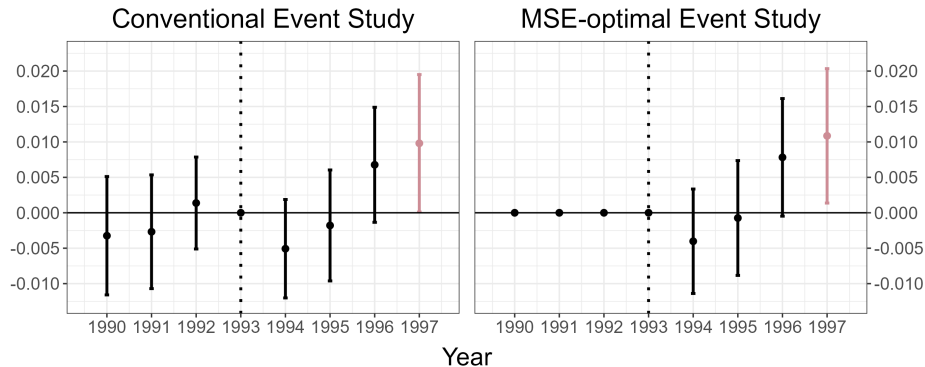


Figure : Estimated effects of the ERI program on math test scores

- For the target parameter in 1997, the MSE-optimal specification yields more precise evidence of a positive treatment effect.